7 A small coil C has 64 turns and cross-sectional area 0.71 cm^2. The coil is placed inside a solenoid as shown in Fig. 7.1.

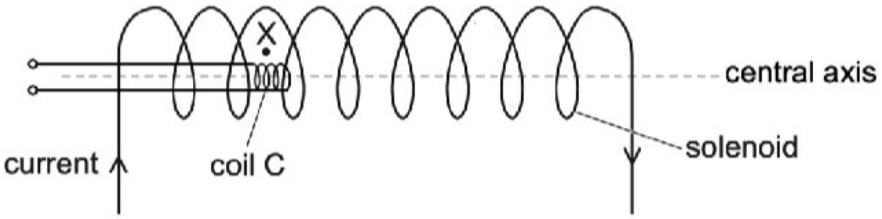


Fig. 7.1 (not to scale)

The centre of coil C is on the central axis of the solenoid.

(a) Coil C is held stationary at X. The current in the solenoid varies so that the magnetic flux density *B* at X varies from time 0 to time 4*t* as shown in Fig. 7.2.

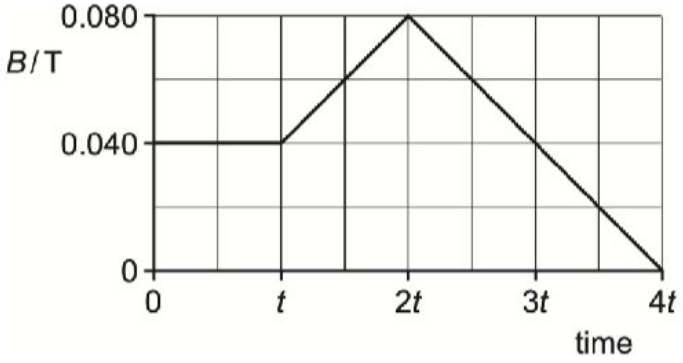


Fig. 7.2

(i) Calculate the maximum magnetic flux linkage in coil C.

flux linkage = ... Wb [2]

(ii) On Fig. 7.3, sketch a line to show the induced electromotive force (e.m.f.) E in coil C from time 0 to time $4t$.

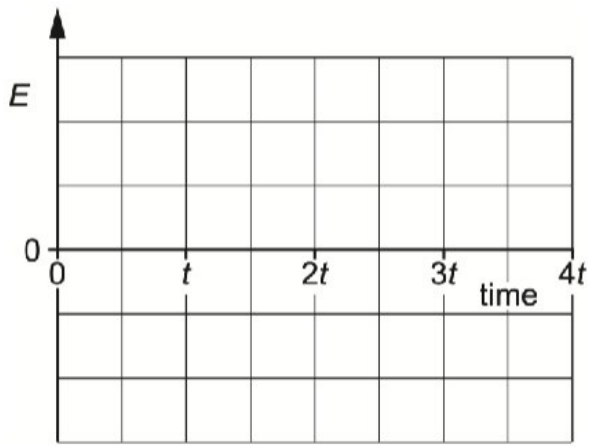


Fig. 7.3

[2]

(b) The terminals of coil C are connected together. Coil C is then removed from inside the solenoid.

With reference to the magnetic fields of the solenoid, explain why a force is required to remove coil C from the solenoid.

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..... [3]

[Total: 7]